



CV2090
Probability & Statistics for
Civil Engineers
(Jul – Nov 2026)

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L10
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Messages from the Last Class

- **Message 1:** Cumulative distribution function (CDF)

$$F_X(x) = \Pr(X \leq x)$$

Properties: (1) ↗

$$(2) F_X(-\infty) = 0$$

$$F_X(+\infty) = 1$$

$$(3) x < y, F_X(x) \leq F_X(y)$$

Non-decreasing function

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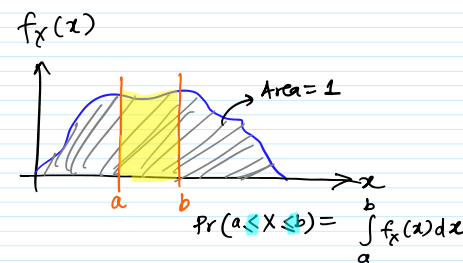
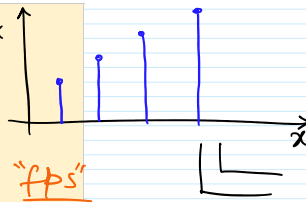
Continuous Random Variables

Probability Density Function (PDF)

$$\text{Temp} = 28^\circ\text{C} \pm 0.5^\circ\text{C}$$

28.0 ... ∞ → Never happens

$$28.4 \pm 0.1^\circ\text{C}$$



Discrete RV

$$\text{PMF, } p_X(x) = \Pr(X=x)$$

$$\text{Properties, } 0 \leq p_X(x) \leq 1$$

$$\sum_x p_X(x) = 1$$

Continuous RV

$$\text{PDF, } f_X(x) = \Pr(X=x)$$

density

$$f_X(x) \geq 0$$

$$\int_{-\infty}^{\infty} f_X(x) dx = 1$$

→ $\Pr(X=x) = 0$
 ↑ for cent.
 → Never check for equality of two real numbers

$$\Pr(a < X < b)$$

$$= \int_a^b f_X(x) dx$$

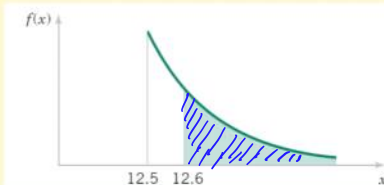
PDF: Class Problem

- Let L = the length of a construction steel bar
- Production target length is 12.5 m. However, the length of the steel bars is affected by disturbances in the extrusion process

- From available data, the PDF can be described as

$$f_X(x) = 20e^{-20(x-12.5)}, \quad x \geq 12.5$$

If a piece with a length longer than 12.6 m is rejected, what is the proportion of pieces rejected?



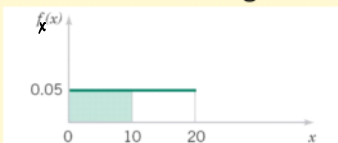
Cumulative Distribution Function (CDF)

	CDF (disc. rv)	CDF (cont. rv)
Definition	$F_X(x) = \Pr(X \leq x)$ $= \sum_{x \leq x_i} p_X(x_i)$	$F_X(x) = \Pr(X \leq x)$ $= \int_{-\infty}^x f_X(x) dx$
Properties	$F_X(-\infty) = 0 \longrightarrow$ $F_X(+\infty) = 1$	$F_X(-\infty) = 0$ $F_X(+\infty) = 1$
	if $x \leq y$, then $F_X(x) \leq F_X(y)$. Non-decreasing function	$\longrightarrow$ same
Rel. with PMF/PDF	$p_X(x_i) = F_X(x_{i+1}) - F_X(x_i)$ $\hookrightarrow$ PMF	$f_X = \frac{dF_X}{dx}$

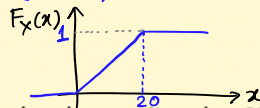
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CDF: Class Problems

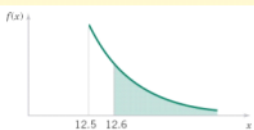
- For concrete strength test experiment, determine the CDF.



$$F_X(x) = \begin{cases} 0, & x < 0 \\ x/20, & 0 \leq x < 20 \\ 1, & x \geq 20 \end{cases}$$



- For the construction steel bar example, determine the CDF.



$$f_X(x) = 20e^{-20(x-12.5)}, \quad x \geq 12.5$$

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Mean, Variance (Expectations)

Mean

Q) Long-term average on a die roll?

3.5

Q) even numbers $\equiv 3 \times$ likely odd numbers
Long-term average?

mean of X , μ_X

$$\mu_X = \sum x_i \overset{\text{arrow}}{p_X(x_i)} = \sum x_i p_X(x_i)$$

$$= \int_{-\infty}^{\infty} x f_X(x) dx = E[X]$$

Expected value

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Variance

- Which one is larger?

Average of squares $E[X^2]$
or ∇

Square of averages $E^2[X]$

Measure of spread

$$\sigma^2 =$$

Variance

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